

* Min Cost of Connecting Ropes : $\rightarrow$ $\rightarrow$ GfG $\rightarrow$ Naukri 360

Cost should be minimum : $\rightarrow$

Sort $\rightarrow (n \log n)$ $2, 3, 4, 6$ $2+3=5$ $Cost=5$

Sort $\rightarrow (n \log n)$ $[5, 4, 6]$ $4, 5, 6$ $4+5=9$ $Cost=9$

Sort $\rightarrow (n \log n)$ $9, 6$ $6, 9$ $6+9=15$ $Cost=15$

empty

$\{n \log n + n \log n + n \log n\}$ very slow.

(Optimized) $O(1)$

(Heap) (min cost)

$\{4, 3, 2, 6\}$ ($L > N < R$) min-heap

$l_1 = \text{minheap.poll}() \Rightarrow 2$

$l_2 = \text{minheap.poll}() \Rightarrow 3$

cost = $2+3=5$ $+ 9 = 14$ $+ 15 = 29$

$l_1 = \text{mt.poll}() \Rightarrow 4$

$l_2 = \text{mt.poll}() \Rightarrow 5$

$4+5=9$

$l_1 = \text{mt.poll}() \Rightarrow 6$

$l_2 = \text{mt.poll}() \Rightarrow 9$

cost = $6+9=15$

removes $O(1)$ $n \log n$

Permutations : $\rightarrow [1, 2, 3]$

How many valid combinations of this array of elements can you achieve without duplicates?

Logic: temp = [], used = [F F F]

S1: Pick 1 No more left: backtrack: Remove 3

temp = [1 2] used = [T T F] Backtrack: Remove 2

S2: Pick 2 temp = [1 2] used = [T T F] Backtrack: Remove 2

temp = [1 2 3] * used = [T T T] * $\rightarrow$ add to result

As soon as you get all combinations of one number, backtrack & remove that number.

temp = [1 3] used = [T F T]

temp = [1 3 2] * used = [T T T] * $\rightarrow$ add to result

Output: [1, 2, 3], [1, 3, 2], [2, 1, 3], [2, 3, 1], [3, 1, 2], [3, 2, 1]

Introduction to Graphs : $\rightarrow$ (Non-linear Data Structures)

It is a non-linear data structure having entities called nodes. Each node is connected to other nodes using "edges".

Depending on the connectivity graphs can be of many types : $\rightarrow$ Nodes/Vertex

Edge $u \leftrightarrow v$ both dir's

1, 2, 3, 4 $\rightarrow$ Nodes

edge

* No direction for the edge = Undirected graph

edge has a direction (Directed graph)

Graph has direction has edge values (weights) $\rightarrow$ Directed Weighted graph

Graph has no direction has edge values (weights) $\rightarrow$ Undirected Weighted graph

Undirected Cyclic graph

From any node if you can travel the graph & come back to the same node, it is cyclic or else it is acyclic

Directed Cyclic graph

Directed Acyclic graph (DAG)

Connected Component 1

Single component graph

Disconnected Component 1

Component 2

Component 3

We can never reach all other nodes from a given node. So, it is called disconnected.

Bipartite graph : $\rightarrow$

Two adjacent nodes have same colour or property Not a bipartite graph

No Two adjacent nodes have same colour or property Bipartite graph

* Directed or Undirected

* Connected or Disconnected

* Weighted or Unweighted

* Cyclic or Acyclic

* Bipartite or Non-Bipartite

Combinations of any of these *** [DAG]

Types of Combinations of graphs

Graph Representations : $\rightarrow$ (Node to itself traversal is not considered)

Adjacency Matrix

Adjacency Matrix

Node List of Neighbours

1 2, 3, 4

2 5

3 2, 3

4 5

5

$n=5, m=7$

Adjacency Matrix

Adjacency List

Node List of Neighbours

1 2, 3, 4

2 5

3 2, 3

4 5

5

* How many of you can confidently say that these 13 days of training skills has at least 5%?

(14, 15) Java + DSA

(Web Applications)

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